

PANDA: Exact Full- C Selected Normalization for Transducer ASR

Lewis Lin

rl4921085@ntu.edu.tw

GitHub repository: <https://github.com/Lewis-panda/panda-selected-normalizer>

ABSTRACT

We use RNN-T/transducer to denote the transducer training formulation with an encoder, prediction network, joint network, and dynamic-programming loss, not to require a recurrent acoustic encoder. This paper studies a systems boundary in that formulation: exact full- C joint-output normalization at fixed full or retained lattice sites. The dynamic program consumes only selected blank and transcript-target log-probabilities, while exact training still requires the softmax denominator over all C output classes.

We present PANDA, an exact full- C selected-normalization method for standard transducer blank/target DP objectives. PANDA streams the full- C log-sum-exp denominator over vocabulary tiles, emits the selected target and blank log-probabilities consumed by the DP, and recomputes tile-local logits in backward to accumulate hidden, weight, and bias gradients. This avoids persistent full $[N, C]$ logits and persistent full `grad_logits` while preserving the fixed-site full- C objective.

On the same-machine exact full- C C-scaling benchmark, the materialized full-logit path first OOMs at $C = 8192$, while PANDA completes through $C = 65536$ without persistent full logits or full `grad_logits`. Retained-lattice rows show composition with pruning, which reduces lattice sites N rather than per-site C materialization. Bounded AISHELL $C = 4336$ and secondary WenetSpeech $C = 5537$ routes show the same boundary entering real transducer training-loop code with lower reported PyTorch peak allocated/reserved memory. Larger- C rows isolate the output-class memory/OOM mechanism; they are not ASR-quality evidence.

1 INTRODUCTION

We use RNN-T/transducer to denote the transducer training formulation with an encoder, prediction network, joint network, and dynamic-programming loss (Graves, 2012), not to require a recurrent acoustic encoder. Modern transducer ASR commonly pairs this formulation with non-recurrent encoders; for example, Transformer Transducer combines Transformer encoders with RNN-T loss (Zhang et al., 2020). End-to-end ASR systems also treat CTC, attention-based encoder-decoder (AED), and transducer decoders as distinct objective/decoder families that can share an encoder (Sudo et al., 2024). This paper reports the standard transducer instance of an exact full- C selected-emission structured-loss normalization/backward boundary; it targets the joint-output normalization interface, not the acoustic encoder architecture.

The motivation is narrower than a survey of ASR objectives. CTC supports streaming but uses stronger conditional-independence assumptions. AED models provide strong sequence modeling, but streaming use requires additional constraints. Transducer objectives combine monotonic streaming alignment with output-history conditioning. In that standard blank/target DP formulation, large C is a plausible design space, but it should be treated here as a systems target rather than as an ASR accuracy conclusion.

At each consumed lattice site, the joiner produces a hid-

den vector and an output projection over C classes. The conventional exact path materializes full logits, normalizes over all C classes, and then passes only two selected emissions to the dynamic program: the blank emission and the transcript-target emission. By selected normalization, we mean this boundary: the DP receives only selected blank and transcript-target log-probabilities, while the normalizer remains the exact full- C softmax denominator over all output classes. As C grows, persistent full $[N, C]$ logits and persistent full `grad_logits` become a concrete memory/OOM barrier to exact full- C experiments.

PANDA exploits this selected-emission/full-denominator structure. It scans the vocabulary dimension in tiles, streams a numerically stable full- C log-sum-exp, records the known target and blank scores, and emits compact target/blank log-probabilities for the transducer DP. In backward, compact DP adjoints for target and blank are mapped back through tiled recomputation to hidden states, vocabulary weights, and bias, avoiding persistent full `grad_logits`. This is the selected-emission structured-loss asymmetry: the structured loss consumes only a small selected set at each site, but the exact denominator and the induced softmax gradient still range over all C output classes. The non-selected classes are therefore still present in the full-softmax derivative; PANDA accumulates their contribution tile-wise instead of storing the full $[N, C]$ adjoint. The same interface applies to a full lattice or to the retained sites chosen by an external Pruned

Conceptual schematic (not a measurement). C = output vocabulary / class count; N = consumed lattice sites (full or retained).

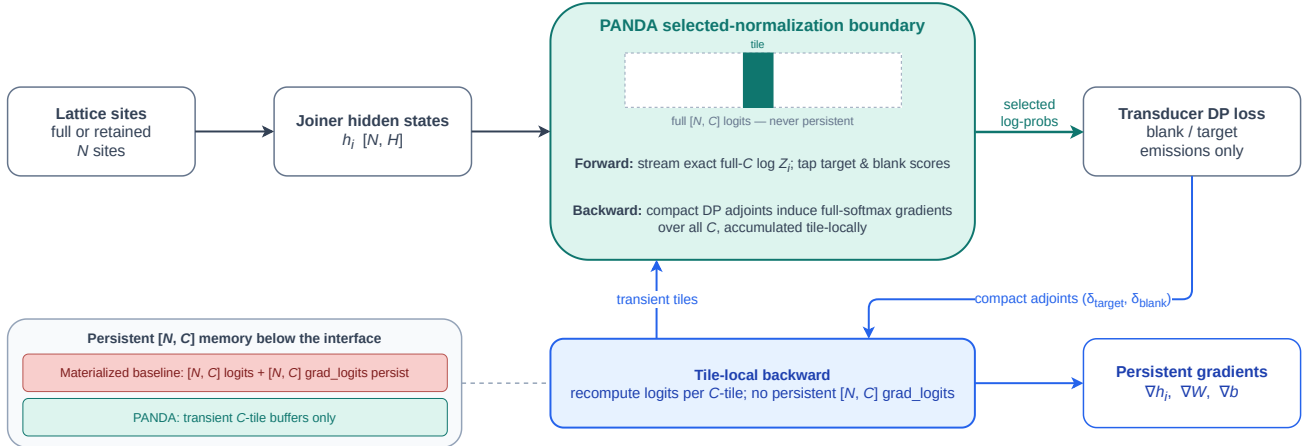


Figure 1. PANDA architecture overview. PANDA sits at the selected-normalization boundary between fixed full or retained transducer sites and the standard blank/target DP. It exposes compact selected log-probabilities and compact DP adjoints while implementations and baselines below the interface differ in whether they persist full $[N, C]$ logits or full `grad_logits`. This is a systems architecture schematic, not a measurement; measured memory/OOM rows and scopes are reported in the evaluation tables.

RNN-T procedure, provided the standard blank/target DP boundary is fixed.

This is a systems contribution in the lineage of materialization-avoidance methods such as online softmax (Milakov & Gimelshein, 2018), FlashAttention-style exact tiling (Dao et al., 2022), and Cut-Cross-Entropy/Liger-style full-vocabulary kernels (Wijmans et al., 2025; Hsu et al., 2024). The novelty is not the tiling primitive. It is the exact full- C selected-emission backward boundary for a structured DP loss: compact selected-emission adjoints induce full- C output-head gradients over all classes, which PANDA accumulates tile-locally without persistent $[N, C]$ logits or `grad_logits`. This boundary was not found in the audited prior work. Pruned RNN-T/k2/fast_rnnt reduce retained lattice sites N but public paths consume retained full- C logits (Kuang et al., 2022); sample-wise transducer training changes the batch/sample scheduling axis (Braun et al., 2023); BAT reduces boundary positions (An et al., 2023); sampled-softmax RNN-T changes the denominator with `K_sample` (Lee et al., 2022); and CCE/Liger/FlashAttention are independent-CE or attention precedents without the audited transducer DP adjoint boundary. PANDA removes one memory/OOM barrier to exact full- C systems exploration without changing the acoustic encoder.

The evidence separates mechanism from workload validation. We evaluate PANDA with a same-machine exact full- C C-scaling/OOM benchmark, retained-lattice composition rows, and two bounded real `training_loop` routes. The direct baseline is same-shape materialized full logits

and full `grad_logits`; the headline evidence is memory/OOM feasibility under fixed full or retained sites. The bounded AISHELL and WenetSpeech routes test whether the same selected-normalization boundary enters real transducer compute-loss/backward/update contracts.

Contributions. This paper makes five scoped contributions.

1. We identify persistent full $[N, C]$ logits and persistent full `grad_logits` as a concrete systems barrier to exact full- C transducer experiments at fixed full or retained lattice sites.
2. We formulate exact full- C selected-emission normalization for structured-loss sites, with standard transducer blank/target DP as the reported instance: the DP consumes selected emissions, while exactness still requires a full- C denominator.
3. We define an interface-level forward/backward boundary that uses tiling as an implementation tool, avoids persistent full logits and full `grad_logits`, and gives a fixed-site exactness argument showing how compact selected-emission DP adjoints induce full-softmax gradient contributions over all C classes.
4. We position the comparison stack around same-shape materialized full-logit baselines, retained-lattice pruning as orthogonal composition evidence, external exact full-logits context, and sampled softmax only as approximate-denominator context with `K_sample`.

5. We report a claim-bounded evidence package whose headline is exact full- C C-scaling/OOM behavior, with AISHELL H1 and secondary WenetSpeech bounded routes as real-route training-loop sanity checks.

2 BACKGROUND AND MOTIVATION

2.1 Transducer Scope and Local Emissions

Following the RNN-T/transducer objective (Graves, 2012), for a lattice site i , the model combines an encoder state and a prediction-network state through a joint network, producing a hidden vector $h_i \in \mathbb{R}^H$. The output projection then forms $z_i = Wh_i + b$, where $W \in \mathbb{R}^{C \times H}$ and C is the output class count. The local blank and target log-probabilities are

$$\log p_i(\emptyset) = z_{i,\emptyset} - \log \sum_{c=1}^C \exp z_{i,c}, \quad (1)$$

$$\log p_i(y_i) = z_{i,y_i} - \log \sum_{c=1}^C \exp z_{i,c}. \quad (2)$$

The target y_i is known from the transcript and lattice position; the blank class is fixed. The denominator, however, must include all C classes. Thus the standard blank/target transducer DP consumes selected emissions, while the local normalizer remains an exact full- C softmax denominator.

The conventional implementation materializes z_i for many sites, giving a full logits tensor with size proportional to NC , where N is the number of full or retained lattice sites. Backward can also expose an equivalent full-grad_logits pressure. This is the memory term PANDA removes.

This scope is deliberately an interface boundary. PANDA targets standard transducer objectives whose DP contract consumes blank and transcript-target emissions from a full- C local normalizer. Hybrid CTC/RNNT systems are in scope only for their RNNT/transducer branch; CTC-only objectives are outside the present claim. Multi-blank Transducer and TDT-style variants are related transducer-family directions, but their selected-emission sets and DP contracts can differ and require separate adaptation and validation.

2.2 Why the Vocabulary Axis Is a Systems Barrier

Large C is a plausible RNN-T design dimension, not an accuracy claim. Character-level Mandarin and multilingual ASR can involve thousands to tens of thousands of output classes. Sampled-softmax RNN-T work reports AISHELL-scale character vocabularies around four thousand classes (Lee et al., 2022), WenetSpeech baselines use thousands of Mandarin characters plus English letters (Zhang et al., 2022), and multilingual RNN-T studies report shared vocabularies in the 11k–19k range (Tjandra et al., 2023).

Table 1. Vocabulary scales motivating exact full- C systems stress tests.

Setting	Vocabulary scale
AISHELL-1 sampled-softmax RNN-T	$\approx 4.2k$ chars
WenetSpeech Mandarin baseline	5.5k+ chars
70-language shared characters	11.3k tokens
70-language char+subword	18.7k tokens

These rows motivate large- C as a plausible RNN-T design space; they do not support accuracy-gain conclusions from increasing C or provide ASR-result evidence at large C .

These references motivate a systems stress regime for exact full- C normalization. They do not support an accuracy-gain conclusion from increasing C , and our synthetic- C rows should not be read as ASR-quality evidence. The question is narrower: if researchers want to explore exact full- C RNN-T variants at fixed full or retained lattice sites, can the implementation remove the persistent full-logits and full-grad_logits memory/OOM barrier?

2.3 Orthogonal Prior-Work Axes

The direct baseline for this paper is not Pruned RNN-T alone. It is the same-shape materialized exact path that stores full logits and exposes full grad_logits pressure for the same fixed full or retained lattice sites. This baseline isolates the systems term PANDA removes.

Sampled-softmax RNN-T (Lee et al., 2022) reduces the vocabulary cost by using a sampled denominator. It is related, but the sampled count should be denoted `K_sample`, and the objective is approximate rather than exact full- C . We therefore keep sampled softmax out of the exact full- C main tables.

Pruned RNN-T and fast-rnnt-style methods reduce the number of retained lattice sites (Kuang et al., 2022). They address N . PANDA addresses per-site C materialization. The methods are complementary, and retained lattice rows in this paper are composition evidence rather than a claim that PANDA replaces pruning.

Cut-Cross-Entropy and Liger-style kernels (Wijmans et al., 2025; Hsu et al., 2024) are closest in systems spirit: tile over a large vocabulary and avoid materializing a large intermediate table. Our claim is not generic cross-entropy novelty. The RNN-T-specific claim is the exact full- C selected-normalization boundary through the target/blank DP interface and nonlinear joiner backward.

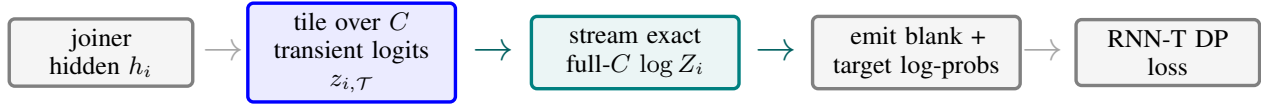
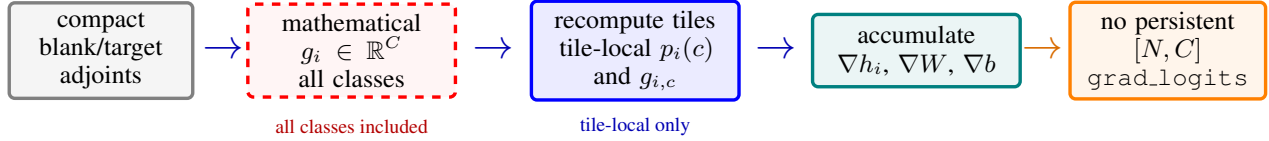
Forward: selected outputs, exact full- C denominator**Backward: compact DP adjoints induce all- C softmax terms**

Figure 2. **Forward/backward boundary for exact selected normalization.** In forward, PANDA streams the exact full- C denominator and emits only blank/target log-probabilities to the transducer DP. In backward, compact DP adjoints induce mathematical full-softmax gradients over all C classes, but PANDA forms only tile-local entries transiently and accumulates `grad_hidden`, `grad_weight`, and `grad_bias`. This is a mechanism schematic, not a new measurement.

3 PANDA SELECTED NORMALIZATION

3.1 Selected-Emission Boundary Versus Tiled Softmax

PANDA uses exact streaming normalization, but it does not claim online softmax or vocabulary tiling as a new primitive. Online softmax and FlashAttention-style implementations provide generic exact tiling and streaming-normalizer patterns (Milakov & Gimelshein, 2018; Dao et al., 2022). Cut-Cross-Entropy and Liger-style kernels address full-vocabulary materialization for independent cross-entropy positions (Wijmans et al., 2025; Hsu et al., 2024). Those systems are important precedents, but the audited boundary in this paper is different: a structured DP loss receives compact selected emissions, while the exact denominator and output-head backward still range over all C classes.

The same boundary was not found in the audited prior work. Pruned RNN-T/k2/fast_rnnt reduce retained lattice sites N , but public paths consume retained full- C logits shaped like $[B, T, s_{\text{range}}, C]$ and rely on log-sum-exp/autograd over those retained logits (Kuang et al., 2022). Sample-wise transducer training reduces the batch/sample scheduling axis while still defining per-sample full- C output scores and output-score gradients (Braun et al., 2023). BAT reduces boundary positions rather than the exact full- C selected-normalization boundary (An et al., 2023). Sampled-softmax RNN-T changes the denominator with `K_sample` (Lee et al., 2022). The PANDA contribution is the selected-emission structured-loss boundary: compact selected-emission adjoints are mapped through the full-softmax derivative and nonlinear output-head backward, then accumulated tile-locally without persistent $[N, C]$ logits or `grad_logits`. This paper reports the standard trans-

ducer blank/target instance; CCE, Liger, FlashAttention, and related systems are treated as neighboring precedents for different boundaries rather than as negative baselines.

3.2 Forward State

PANDA replaces persistent full-logit materialization with a streaming state per site. For a vocabulary tile $\mathcal{T}$, it computes tile-local logits $z_{i,c}$ only transiently and updates

$$m'_i = \max(m_i, \max_{c \in \mathcal{T}} z_{i,c}), \quad (3)$$

$$\ell'_i = \ell_i \exp(m_i - m'_i) + \sum_{c \in \mathcal{T}} \exp(z_{i,c} - m'_i). \quad (4)$$

After all tiles, $\log Z_i = m_i + \log \ell'_i$ is the exact full- C log-sum-exp denominator. For the reported standard RNN-T instance, the selected set is the blank and the valid transcript target. During the same scan, PANDA records z_{i,y_i} and $z_{i,\emptyset}$ when the tile contains the known target or blank class. The DP receives only

$$\log p_i(y_i) = z_{i,y_i} - \log Z_i, \quad (5)$$

$$\log p_i(\emptyset) = z_{i,\emptyset} - \log Z_i. \quad (6)$$

This is exact selected normalization: selected outputs, but a full- C denominator. It is not sampled softmax, and it does not replace the RNN-T objective with an approximate denominator.

3.3 Backward Boundary

The structured loss returns compact upstreams for selected emissions; in the reported RNN-T instance these are blank and target emissions. PANDA maps those upstreams

Conceptual persistent memory terms

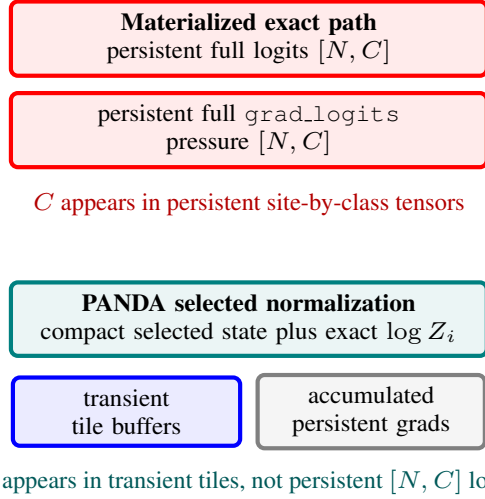


Figure 3. **Persistent memory term removed by PANDA.** The materialized exact path stores persistent full $[N, C]$ logits and exposes full `grad_logits` pressure at fixed sites. PANDA keeps compact selected emissions and $\log Z_i$ state, while full- C quantities are recomputed in tile-local transient buffers. This conceptual diagram explains the memory/OOM mechanism; measured PyTorch peak allocated/reserved MiB values are reported in the evaluation tables.

through the full-softmax derivative without storing a full $[N, C]$ gradient tensor. Backward recomputes each tile’s logits from the saved compact forward state, uses $\log Z_i$ to recover tile-local softmax probabilities, and accumulates only the gradients that must persist:

$$\nabla h_i, \quad \nabla W, \quad \nabla b.$$

Tile-local logits and tile-local gradient logits are discarded after accumulation.

Proposition 1: fixed-site selected-set exactness. Consider fixed consumed sites i and a scalar structured loss L whose local softmax interface consumes a selected set $\mathcal{S}_i \subseteq \{1, \dots, C\}$. Let the output projection logits be $z_{i,c} = W_c h_i + b_c$, with exact local normalizer $Z_i = \sum_{c=1}^C \exp z_{i,c}$ and $p_i(c) = \exp z_{i,c} / Z_i$. The loss receives $r_{i,s} = \log p_i(s) = z_{i,s} - \log Z_i$ for $s \in \mathcal{S}_i$ and backward returns compact adjoints $a_{i,s} = \partial L / \partial r_{i,s}$. In the standard RNN-T instance reported here, $\mathcal{S}_i = \{\emptyset, y_i\}$ when the target emission is valid; transcript targets exclude blank, so a valid target and blank are distinct. If the target emission is invalid, y_i is omitted from $\mathcal{S}_i$, equivalently the target adjoint is zero. Then the mathematical full-logit adjoint for any

class c is

$$g_{i,c} \equiv \frac{\partial L}{\partial z_{i,c}} = \sum_{s \in \mathcal{S}_i} a_{i,s} (1[c = s] - p_i(c)) \quad (7)$$

$$= a_{i,c}^{\text{sel}} - p_i(c) \sum_{s \in \mathcal{S}_i} a_{i,s}, \quad (8)$$

where $a_{i,c}^{\text{sel}} = a_{i,c}$ if $c \in \mathcal{S}_i$ and 0 otherwise. Thus non-selected classes are not ignored: they receive the denominator term induced by the selected-emission adjoints.

Sketch. Since $r_{i,s} = z_{i,s} - \log \sum_{c=1}^C \exp z_{i,c}$, differentiating $r_{i,s}$ with respect to $z_{i,c}$ gives $1[c = s] - p_i(c)$ and summing over the compact DP adjoints yields Eq. 7. A materialized implementation can form the full vector $g_i \in \mathbb{R}^C$ for every site. PANDA instead scans classes in tiles, recomputes $p_i(c)$ from the transient tile logits and saved $\log Z_i$, forms tile-local entries of $g_{i,c}$ only transiently, and accumulates

$$\nabla W_c += \sum_i g_{i,c} h_i^\top, \quad (9)$$

$$\nabla b_c += \sum_i g_{i,c}, \quad (10)$$

$$\nabla h_i += \sum_c g_{i,c} W_c. \quad (11)$$

Thus g_i exists mathematically and has length C , but PANDA never stores the collection of these vectors as persistent $[N, C]$ `grad_logits`. The resulting gradients match the fixed-site materialized full-softmax computation up to floating-point reduction order.

Appendix Table 6 reports quantitative fixed-site checks: cached-hidden rows match target/blank/ $\log Z$ with maximum absolute error 0.0, and the large- C compact-DP backward row has worst reported output-head gradient relative ℓ_2 error below 8.7×10^{-7} with cosine at least 0.99999999999987754. The large- C compact-DP row is backward-parity evidence only because forward target/blank/ $\log Z$ parity was not stored in that artifact. The table also includes a synthetic $|\mathcal{S}_i| > 2$ multi-selected parity row, showing that the selected-set algebra/backward boundary is not limited to the blank/target two-emission RNN-T instance.

3.4 Exactness Scope

The exactness statement is conditional on fixed consumed sites and a fixed selected-emission contract $\mathcal{S}_i$. Mathematically, Eq. 7 holds for any selected set $\mathcal{S}_i$ that receives exact full- C log-probabilities from the local softmax. The implemented and evaluated instance in this paper is the standard blank/target transducer boundary. Given the same model parameters and the same full or retained RNN-T lattice

sites, PANDA computes the same local full- C denominator and the same selected blank/target log-probabilities up to floating-point reduction order. In a pruned setup, the retained sites are supplied by an external pruning method; PANDA does not claim to replace the pruning algorithm. Multi-blank Transducer, TDT-style objectives, or other structured losses would require a separate selected-emission contract and validation before they could be claimed as supported implementations.

4 IMPLEMENTATION

4.1 Triton Boundary

The current implementation evidence centers on Triton selected-normalizer kernels (Tillet et al., 2019). The kernels tile over site count N , vocabulary size C , and hidden dimension H . The flat forward wrapper uses blocks such as `block_n=32`, `block_c=256`, and `block_h=32`; the flat backward wrapper uses smaller vocabulary tiles for gradient accumulation. The forward split path stores compact partial max/sum/target/blank states and reduces them to final `target_logp[N]`, `blank_logp[N]`, and `logZ[N]`. It does not materialize $[N, C]$ logits.

Backward recomputes tile-local logits from `hidden_sites`, W , b , `logZ`, and compact `dtarget/dblank` upstreams. It accumulates `grad_hidden`, `grad_weight`, and `grad_bias`.

A fixed-stack profiler artifact now covers this selected-normalizer kernel boundary. The fixed stack reports driver 595.71.05, CUDA 13.2, PyTorch 2.10.0+cu130, ncu 2026.1.1.0, and nsys 2025.6.3.541. Its `profile_scope` is `selected_normalizer_kernel_boundary`, and its `memory_scope` is `selected_normalizer_smoke`. `torch.profiler` recovered CUDA selected-normalizer rows for `tiny_selected_smoke` ($N = 128, H = 64, C = 512$), `aishell_native_c4336_r20` ($N = 32093, H = 512, C = 4336$), and `integrated_c8192_r20` ($N = 32093, H = 512, C = 8192$). For each profiled shape, the profiler-derived selected-normalizer launch count is three: `_selected_hidden_partial_kernel`, `_selected_hidden_reduce_kernel`, and `_selected_hidden_backward_atomic_kernel`.

In the selected-normalizer smoke memory scope, the $C = 4336$ shape reports CUDA event time 102.118 ms and PyTorch peak allocated/reserved memory 143.192/194.000 MiB; the $C = 8192$ shape reports 183.168 ms and 158.284/194.000 MiB. ncu register, shared-memory, and occupancy fields are available only for the tiny selected-normalizer smoke: the partial kernel reports 152 registers/thread, 73728 B dynamic shared memory/block, and

8.33%/8.32% theoretical/achieved occupancy; the reduce kernel reports 26 registers/thread, 0 B dynamic shared memory/block, and 100%/8.26%; the backward atomic kernel reports 128 registers/thread, 40960 B dynamic shared memory/block, and 16.67%/8.33%. Spill metrics were not reported by the selected ncu sections. These measurements are kernel-boundary facts only; they are not training-loop, production/default-backend, speedup, or multi-hardware evidence, and the tiny ncu metrics are not generalized to the full-shape rows.

4.2 Bucketed Launch Sites

The AISHELL guarded helper pads selected-normalizer launch sites to power-of-two buckets. Hidden padding is zero, padded target ids are -1 , and padded upstream gradients are zero. Compact DP tensors, tokenizer/head semantics, native AISHELL $C = 4336$, and blank id 2 are unchanged.

This bucketed- N strategy addresses a Triton dynamic-shape JIT problem. The current timing characterization comes from the bucketed guarded AISHELL H1 bounded `training_loop` row, where the guarded path is near parity but slightly slower; the memory/OOM benefit remains the headline. Archived dynamic- N or cross-artifact timing ratios are not used as the current speed characterization.

4.3 Guarded k2_mi Compact-DP Candidate

The guarded `k2_mi` route is an optional runtime candidate, not a production/default backend. It uses selected-normalizer outputs as compact target/blank scores, calls a k2 mutual-information recursion as the DP leaf, maps compact DP gradients back to target/blank upstreams, and invokes selected-normalizer backward for hidden sites, output weights, and bias; the k2/fast-rnn lineage is cited only as the pruned-transducer software context (Kuang et al., 2022).

An implementation feasibility result shows that the legacy backward helper can be wrapped in a `torch.autograd.Function`. The synthetic one-batch feasibility smoke returned a differentiable loss, matched the old manual loss exactly (18.897930145263672), produced finite gradients, and matched manual encoder/decoder/weight/bias gradients. The AISHELL H1 official-DataModule bounded continuation then integrated this boundary, with both materialized and guarded paths completing 5000/5000 steps through the `normal_compute_loss -> loss.backward() -> optimizer.step()` contract.

4.4 Artifact Boundary

The reproducibility package validates existing paper-ready tables and records fixed rerun commands in dry-run form. It does not establish second-hardware replication, production/default backend status, or full training convergence. Remaining engineering work includes fixed-script external reruns and second-GPU rows after local smoke checks are in place.

5 EVALUATION AND MAIN RESULTS

5.1 Claim-Bounded Benchmark Stack

The evaluation separates three claims. **C1, mechanism:** for fixed full or retained lattice sites, PANDA avoids persistent $[N, C]$ logits and `grad_logits` and moves the exact full- C memory/OOM boundary. The controlled C -scaling/OOM sweep is the direct mechanism test because it isolates the output-class axis C , where persistent full-logit materialization scales. **C2, integration:** the same selected-normalization boundary can enter a real transducer `compute_loss` $\rightarrow$ `backward` $\rightarrow$ `optimizer` contract. The AISHELL and WenetSpeech bounded `training_loop` routes test that integration layer at native vocabularies. **C3, workload generalization:** full convergence, formal WER/CER preservation, and large- C multilingual ASR quality are next-stage workload-validation questions rather than current claims.

Pruned retained-site composition tests a different axis of C1: Pruned RNN-T reduces retained sites N , while PANDA reduces per-site C materialization. Matched practical exact full- C `integrated_path` context rows are reported for the same parent harness, environment stack, row policy, and R1 model fixture: all six rows run at $C = 4336$; at $C = 8192$, guarded `k2_mi/PANDA`, `selected_torch`, and NeMo bottom `RNNTLossNumba` run, while materialized full logits, `warprnnt_numba`, and official `SpeechBrain TransducerLoss` hit OOM. These rows are exact full- C systems context rather than a PANDA speed claim.

The three main-result questions are:

1. **C-scaling/OOM:** how far can exact full- C systems rows run before OOM under the same-machine setup when persistent full logits are present or removed?
2. **AISHELL H1 bounded continuation:** in a bounded native $C = 4336$ `training_loop`, does the guarded path lower PyTorch peak allocated/reserved memory while preserving the reported bounded CER sanity check?
3. **Pruned composition:** after an external Pruned RNN-T method reduces retained lattice sites N , does selected normalization still reduce per-site C materialization?

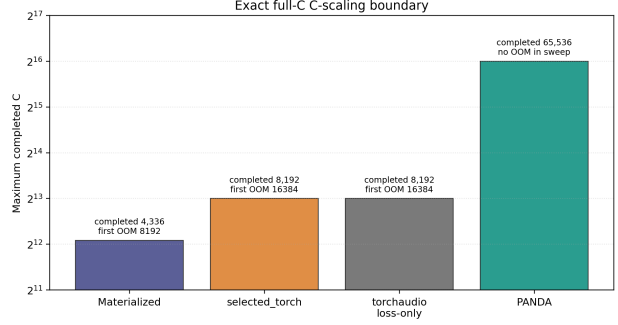


Figure 4. Exact full- C C-scaling boundary. Materialized full logits first OOM at $C = 8192$; guarded `k2_mi/PANDA` completes through $C = 65536$. Scope is `integrated_path` for materialized, `selected_torch`, and guarded `k2_mi/PANDA` rows, and `loss_only` for the `torchaudio` row. The corresponding PyTorch peak allocated/reserved MiB values are reported in Table 2. Synthetic- C rows support the C1 mechanism claim.

Every experimental row declares one scope: `loss_only`, `integrated_path`, or `training_loop`. Memory is reported as PyTorch peak allocated/reserved MiB. Synthetic- C systems rows support C1 rather than C3. Sampled-softmax related work is kept out of the exact full- C main tables because it uses an approximate denominator and sampled count `K_sample`. Appendix Table 8 adds a secondary WenetSpeech $C = 5537$ bounded subset. It is a real ASR route rather than a synthetic- C row, and it supports the C2 integration layer rather than C3 workload validation.

5.2 Exact Full- C Scaling

Table 2 and Figures 4 and 5 summarize the main systems boundary. Materialized full logits complete only the native $C = 4336$ row and first OOM at $C = 8192$. `selected_torch` first OOMs at $C = 16384$. Guarded `k2_mi/PANDA` completes the tested range through $C = 65536$ while avoiding persistent full logits and full `grad_logits`. This is the headline result: one memory/OOM barrier to exact full- C exploration is removed in this reported systems scope. Here `selected_torch` is an exact selected-normalization reference path implemented with PyTorch-level operations. It removes persistent full logits and full `grad_logits` at the selected interface, but lacks the guarded `k2_mi/PANDA` tiled kernel boundary; it is therefore an implementation ablation rather than an external method. The label “guarded `k2_mi/PANDA`” denotes an implementation stack, not an attribution that the DP recurrence alone removes the full- C memory term: `k2_mi` supplies the compact blank/target DP leaf and recurrence, while PANDA supplies the exact full- C selected-normalization and backward boundary. The memory/OOM

Table 2. Exact full- C C-scaling/OOM boundary. Systems rows come from fixed full-mode artifacts ingested read-only by the reproducibility package. Memory is PyTorch peak allocated/reserved MiB at the listed boundary row; OOM peaks are failure-attempt peaks. Synthetic- C rows are not ASR-quality rows. The torchaudio row (Yang et al., 2021) is a loss/backward backend after full logits already exist; its timing is not full-update timing.

Path	Scope	Max completed C	First OOM C	Boundary PyTorch peak allocated/reserved MiB	Persistent full logits	Persistent full grad logits
Materialized full logits	integrated.path	4,336	8192	14206.135/14714.000 at first OOM	Yes	Yes (eff.)
selected.torch	integrated.path	8,192	16384	15120.877/15624.000 at first OOM	No	No
guarded k2.mi / PANDA	integrated.path	65,536	none	6121.012/8008.000 at $C = 65536$	No	No
torchaudio.functional.rnn_loss	loss_only	8,192	16384	10939.176/10962.000 at first OOM	Yes	Yes (eff.)

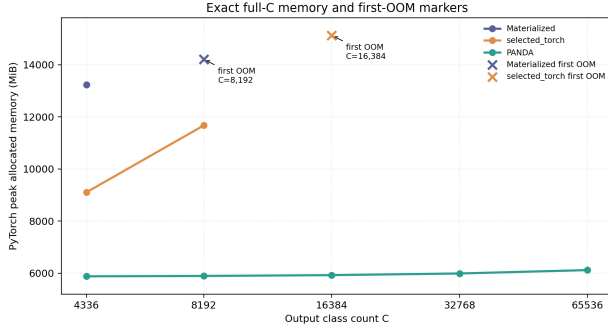


Figure 5. PyTorch peak allocated memory for the exact full- C fixed full-mode sweep. Scope is integrated.path for the plotted materialized, selected.torch, and guarded k2.mi/PANDA rows. The y-axis is PyTorch peak allocated memory in MiB; reserved MiB and row scopes are reported in Table 2. Cross markers denote first-OOM failure attempts. Synthetic- C rows support the C1 mechanism claim.

movement is attributed to removing persistent $[N, C]$ logits and grad_logits through selected normalization plus tile-local full- C recomputation/backward. The torchaudio row (Yang et al., 2021) is a loss_only full-logits backend after logits already exist, so its timing should not be compared as full-update timing against integrated.path or training_loop rows.

5.3 AISHELL H1 Bounded Continuation

The AISHELL native $C = 4336$ H1 bounded continuation row is the current training-loop sanity check. It shows that the guarded path can run through an official-DataModule bounded continuation at native vocabulary size; it is not evidence for large- C ASR results. Both paths use the official DataModule route, the same seed, checkpoint initialization, optimizer/LR, and batch order, and both complete 5000/5000 bounded steps. Guarded k2.mi/PANDA lowers PyTorch training-loop peak allocated/reserved memory from 2293.222/2462.000 MiB to 1222.858/1490.000 MiB. The guarded row is near parity but slightly slower, and it preserves the reported bounded CER sanity check: dev CER 0.048699480 versus 0.048714090, test CER 0.053872954 versus 0.053872954, with dev/test hypothesis mismatches

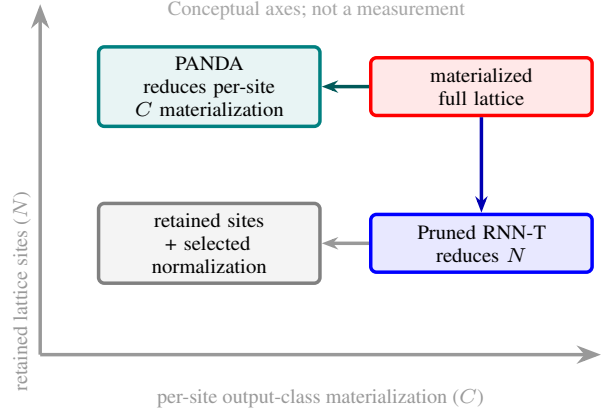


Figure 6. Why pruning and selected normalization compose. Pruned RNN-T reduces the number of retained lattice sites N , while PANDA removes persistent per-site full- C materialization. This is a conceptual axes diagram rather than a measured row; the measured retained-site rows in Table 4 use integrated.path scope and report PyTorch peak allocated/reserved MiB.

2/1. This supports the C2 integration claim; C3 workload validation remains outside the current evidence package.

Secondary WenetSpeech bounded subset. Appendix Table 8 reports a WenetSpeech Zipformer RNN-T/transducer bounded subset at $C = 5537$ and training_loop scope. The guarded k2.mi/PANDA path lowers PyTorch peak allocated/reserved memory from 6040.212/7348.000 MiB to 2119.542/2572.000 MiB and, in this bounded subset, has observed lower mean step time. This partially addresses the coverage concern that the real-ASR route evidence otherwise centers on AISHELL $C = 4336$, but it remains below $C \geq 8192$ and is interpreted as C2 integration evidence.

5.4 Pruned Retained-Site Composition

These rows answer a common baseline question: if a system already uses Pruned RNN-T, does PANDA still remove a separate memory term? The answer is yes under the reported retained-site integrated.path systems setup, because pruning reduces N while PANDA removes retained full logits in the C direction. The larger- C retained rows are C1 systems/OOM rows.

Table 3. AISHELL H1 native $C = 4336$ bounded-continuation sanity. Scope is `training_loop`; both paths use the official DataModule route and complete 5000 bounded continuation steps. Memory is PyTorch peak allocated/reserved MiB. This is not full training or broad WER/CER preservation.

Path	Scope	Steps	Mean/median ms	PyTorch peak allocated/reserved MiB	Dev CER	Test CER	Hyp mismatch dev/test	Full logits
Materialized	<code>training_loop</code>	5000/5000	66.182/66.607	2293.222/2462.000	0.048714090	0.053872954	2/1	Yes
guarded k2.mi / PANDA	<code>training_loop</code>	5000/5000	70.734/71.808	1222.858/1490.000	0.048699480	0.053872954	2/1	No

Table 4. Pruned retained-site composition. Scope is `integrated_path`. Retained full-logits rows consume retained logits $[B, T, r, C]$; PANDA+pruned avoids persistent retained full logits and full grad logits. Memory is PyTorch peak allocated/reserved MiB. Larger- C rows are synthetic systems rows, not ASR-quality evidence.

C	Scope	Retained full PyTorch peak allocated/reserved MiB	PANDA+pruned PyTorch peak allocated/reserved MiB	Alloc ratio
4,336	<code>integrated_path</code>	336.823/370.000	46.118/62.000	7.304x
8,192	<code>integrated_path</code>	602.540/624.000	68.756/90.000	8.763x
16,384	<code>integrated_path</code>	1175.228/1196.000	116.850/140.000	10.058x
32,768	<code>integrated_path</code>	2320.603/2340.000	213.037/234.000	10.893x
65,536	<code>integrated_path</code>	4616.353/4628.000	405.412/426.000	11.387x

External practical exact full- C baselines are reported as appendix context in Table 7. The same-shape materialized full-logits path remains the direct mechanism baseline for the persistent $[N, C]$ term; the external rows provide practical exact full- C `integrated_path` context. Table 7 uses only matched rerun timing from the same harness; older copied-R1 and RelatedWork overlay timings are `cross_artifact_same_scope_context` and are not used as matched timing evidence. The paper’s main interpretation remains the exact full- C memory/OOM boundary rather than PANDA speedup.

6 DISCUSSION

The C-scaling result establishes the main systems consequence of Proposition 1: once full-logit and full-grad-logits materialization is removed from fixed sites, exact full- C transducer experiments cross OOM boundaries that stop the materialized and PyTorch-reference paths. This result isolates the output-class dimension C and explains why PANDA is best read as a memory/OOM feasibility mechanism rather than a new ASR recipe.

The real-route rows establish that the same interface can enter transducer `training_loop` executions. AISHELL H1 provides the primary bounded continuation sanity check at native $C = 4336$, while the WenetSpeech $C = 5537$ bounded subset adds a second transducer route. Together with the retained-site rows, these experiments show that PANDA composes with the standard blank/target DP boundary and with external lattice-site reduction.

The remaining open work is engineering breadth rather than a change in the mathematical claim. Appendix tables record artifact validation, external full-logits context, sampled-softmax caveats, and the non-default retained-site adapter prototype. Section 7 lists the boundaries that are intentionally outside the current claim, including ASR qual-

ity, full-training speedup, multi-hardware generality, and production/default backend status.

Public code. A lightweight implementation package for the selected-normalizer boundary is available at <https://github.com/Lewis-panda/panda-selected-normalizer>. The package contains random-tensor forward/backward smoke tests for the standard blank/target contract and for a synthetic multi-selected contract, and reports parity against a dense full- C oracle while avoiding persistent full logits and persistent full grad-logits. The repository is intended to make the boundary inspectable and runnable; the ASR recipe, training/decode artifacts, and reproduction-package tables remain separate from the lightweight code package.

7 LIMITATIONS

Applicability boundary. PANDA reports the standard blank/target transducer instance of the selected-emission structured-loss boundary. In a hybrid CTC/RNNT system, the claim applies only to the RNNT/transducer branch. CTC-only objectives are outside scope. The selected-set exactness argument is mathematical; Multi-blank Transducer, TDT-style variants, or other structured losses require separate selected-emission contracts and validation before they can be claimed as supported implementations.

Evidence boundary. The current evidence package establishes C1 mechanism evidence and C2 bounded integration evidence. C1 is supported by the C-scaling/OOM and pruned-composition rows; C2 is supported by AISHELL H1 bounded continuation at native $C = 4336$ and the secondary WenetSpeech $C = 5537$ bounded subset. C3 workload validation—full convergence, formal WER/CER preservation, ASR quality improvement, and large- C multilingual ASR-result evidence—remains outside the current claim.

Implementation boundary. Guarded `k2_mi/PANDA` is an optional runtime candidate rather than a production/default backend. The retained-site adapter extension is also explicit opt-in and non-default, with an atomic fall-back available. Kernel profiling is scoped to the selected-normalizer boundary: profiler-derived launch counts cover the fixed-stack boundary shapes, while `ncu` register/shared-memory/occupancy metrics are tiny-smoke-only and should not be read as full-shape or `training_loop` evidence.

Generality and timing boundary. All headline rows are same-machine rows; the artifact package includes a second-GPU runbook, but no second NVIDIA replication has been run. Timing is not a full-training speedup claim, and the practical exact full- C external `integrated.path` rows in Appendix 7 are scoped systems context rather than a PANDA speedup result. They are not mixed with `loss_only` or `training_loop` timing. Sampled softmax remains related work with an approximate denominator `K_sample`; no sampled-softmax superiority or large- C degradation claim is made.

8 CONCLUSION

PANDA targets a specific RNN-T systems bottleneck. Large- C RNN-T is a plausible design space, but exact full- C exploration is blocked by the conventional path’s persistent full $[N, C]$ logits and `full_grad_logits` pressure. By tiling over C , streaming the exact full- C denominator, and recomputing tiles in backward, PANDA preserves the fixed-site objective while removing persistent full-vocabulary materialization.

The current evidence supports a scoped memory/OOM story. Materialized full logits first OOM at $C = 8192$, while guarded `k2_mi/PANDA` completes through $C = 65536$ in the reported same-machine systems sweep. PANDA also composes with retained-lattice pruning: pruning reduces lattice sites N , while PANDA reduces per-site C materialization. The AISHELL $C = 4336$ bounded continuation is a real training-loop sanity check for the guarded path, not ASR-result evidence at large C . The secondary Wenet-Speech $C = 5537$ bounded subset adds a second real-ASR route with lower PyTorch peak allocated/reserved memory for the guarded path, but it remains bounded subset systems evidence rather than formal WER/CER or large- C quality evidence. The public draft and code release therefore separate a reproducible selected-normalizer mechanism from broader ASR-recipe questions. Remaining work includes production/default backend status, full-training speed, multi-hardware replication, broad WER/CER evidence, and ASR-quality effects of changing C ; sampled-softmax comparisons remain separate approximate- denominator questions.

Table 5. Artifact package validation state. Validators pass on existing artifacts and paper-ready tables. The fixed C=4336 smoke wrapper executed successfully, and fixed full-mode artifacts were ingested read-only. Second-GPU replication remains pending.

Check	Value	Status
Required artifacts	20/20	pass
Paper-ready CSV rows	20/15/7/9	pass
Merged rollout rows	49	pass
JSON files parsed	11	pass
JSONL files parsed	2	pass
Repro-package JSONL step rows	1000 (not H1 table)	pass
H1 official step rows	10000	pass
Fixed smoke wrapper	4	fresh_compute_smoke_pass
Fixed full-mode ingest	read_only_existing_artifacts	fresh_compute_full_ingest_pass
Second-GPU replication	pending	not run

A GENERATED TABLES AND ARTIFACT SOURCES

The main generated tables and figures are produced by `scripts/generate_figures_and_tables.py`. The source artifacts are the paper-ready benchmark tables, the AISHELL H1 official-DataModule 5000-step bounded continuation artifact, the WenetSpeech $C = 5537$ bounded subset artifact, the non-default retained-site adapter prototype artifacts, and the reproducibility package.

Table 10. Related-work and context inventory. Sampled-softmax (Lee et al., 2022) uses approximate denominators and `K.sample`. Practical exact full- C external `integrated_path` rows are in Table 7. Other rows cite related methods/toolkits for orientation.

Backend/work	Role	Evidence scope(s)	Exact full- C	Approx. denom.	<code>K.sample</code>	Safe reading
NeMo bottom <code>RNNTLossNumba / warprnt.numba / official</code>	context	<code>integrated_path</code>	Yes	No	–	practical exact full- C rows in Table 7; no speedup claim
SpeechBrain <code>TransducerLoss optimized_transducer</code>	inventory	n/a	n/a	n/a	–	inventory only; no local metric in this table
<code>warp-transducer build path</code>	inventory	n/a	n/a	n/a	–	separate build inventory; <code>warprnt.numba</code> rows in Table 7
Sampled-softmax RNN-T	sampled softmax	<code>training_loop</code>	No	Yes	300/500/1000	sampled-softmax is approximate; keep out of exact full- C main tables
BAT / Boundary-Aware Transducer	context	n/a	n/a	n/a	–	orientation only; runnable exact full- C rows, where present, are in Table 7
<code>warp-transducer / optimized_transducer / SpeechBrain losses</code>	context	n/a	n/a	n/a	–	orientation only; runnable exact full- C rows, where present, are in Table 7
NeMo fused RNNT / <code>SampledRNNTJoint / batch splitting</code>	context	n/a	n/a	n/a	–	orientation only; runnable exact full- C rows, where present, are in Table 7
<code>Sample-wise transducer training</code>	context	n/a	n/a	n/a	–	orientation only; runnable exact full- C rows, where present, are in Table 7
Cut Cross-Entropy	context	n/a	n/a	n/a	–	orientation only; runnable exact full- C rows, where present, are in Table 7

B ENGINEERING EXTENSION: NON-DEFAULT RETAINED-SITE ADAPTER PROTOTYPE

This appendix records plugin/backend-direction evidence for a non-default retained-site adapter. It does not change the main thesis: PANDA remains an exact full- C selected-normalization systems enabler whose core claim is avoiding persistent full logits and persistent full `grad_logits` to improve memory/OOM behavior. The adapter evidence asks a narrower engineering question: can the selected-normalizer boundary be connected to a true retained site layout behind an explicit opt-in shape gate while preserving the compact target/blank interface?

The current answer is a scoped yes for the reported artifacts. A non-default adapter API contract exists for the `true_k2.pruned_retained_sites` layout with flat order `((b*T+t)*r+retained_slot)`. A real retained-site fixture validation passed at $C = 4336$ with `blank_id=0`, retained hidden shape `[4, 173, 5, 512]`, $N = 3460$ flat sites, and 2260 active sites. Subsequent recipe-native checks validated one-batch adapter smoke, one-batch loss/backward smoke, one optimizer-step smoke, a 20-step short smoke, and staged fixed-fixture replay through 100 and 500 steps. In the recipe-native smoke/gate rows, both the atomic baseline and the explicit opt-in `lower_group64_c128h32_w4` candidate keep `persistent_full_logits=false` and `persistent_full_grad_logits=false`.

Table 6. Quantitative fixed-site selected-normalizer correctness and equivalence rollup. These rows summarize existing artifact metrics for Proposition 1. C is the vocabulary size / output class count. Retained-site rows are conditional on the fixed retained sites supplied to the selected normalizer. The PANDA paths avoid persistent $[N, C]$ logits and persistent $[N, C]$ `grad_logits`; full retained baselines, where used, are references/oracles and should not be read as PANDA memory behavior. The table is correctness and adapter-semantics evidence only, not training, decode, WER/CER, ASR-quality, speedup, production/default-backend, or multi-hardware evidence.

Panel A: quantitative metrics.						
Scope / artifact	Layout / C	Reference	Quantity	Max abs error	Relative ℓ_2 error	Cosine
Large- C compact-DP boundary	Compact DP selected-normalizer boundary; $C = 4336, 8192, 16384, 32768, 65536$	Atomic b2c128h32 selected-normalizer backward baseline	Backward: <code>grad_hidden</code> ; <code>grad_weight</code> ; <code>grad_bias</code> . Forward target/blank/log Z parity is not reported here.	0.0: 7.62939453125 $\times 10^{-8}$; 6.103515625 $\times 10^{-4}$	0.0; 4.970971 $\times 10^{-7}$; 8.667931 $\times 10^{-7}$	0.99999999999999978; 0.99999999999997754; 0.99999999999999978
Cached-hidden selected-normalizer boundary	Cached-hidden RNN-Faste flatten order; $C = 4336, 8192, 16384, 32768, 65536$	Artifact-local cached-hidden reference path	Forward: target log-prob, blank log-prob, log Z . Backward: <code>grad_hidden</code> ; <code>grad_weight</code> ; <code>grad_bias</code> .	Forward all 0.0; backward 0.0; 2.0503997802734375 $\times 10^{-5}$; 4.0531158447265625 $\times 10^{-6}$	Backward 0.0; 1.207896 $\times 10^{-6}$; 1.851151 $\times 10^{-6}$	Backward 0.99999988079071045; 0.99999994039535522; 0.9999997615814209
Cached-hidden backend-adapter boundary	Cached-hidden adapter boundary with feature-gradient accounting; $C = 4336, 8192, 16384, 32768, 65536$	Artifact-local atomic adapter reference	Forward: target log-prob, blank log-prob, log Z . Backward: cached-hidden gradient; <code>grad_weight</code> ; <code>grad_bias</code> .	Forward all 0.0; backward 0.0; 2.002716984453125 $\times 10^{-5}$; 6.205740604400635 $\times 10^{-6}$	Backward 0.0; 1.212804 $\times 10^{-6}$; 2.490764 $\times 10^{-6}$	Backward 0.99999988079071045; 0.99999994039535522; 0.9999997615814209
Synthetic multi-selected second-instance parity	Synthetic multi-blank-style selected-set contract; $C = 4336, 8192, S_i = 3.5$	Dense full- C log_softmax autograd oracle	Forward selected log-prob and log Z ; scalar selected-emission loss; backward <code>grad_hidden</code> , <code>grad_weight</code> , and <code>grad_bias</code> .	Selected log-prob 1.907 $\times 10^{-6}$; log Z 1.907 $\times 10^{-6}$; loss 1.907 $\times 10^{-6}$	Loss 5.567 $\times 10^{-7}$; <code>grad_hidden</code> 5.746 $\times 10^{-7}$; <code>grad_weight</code> 2.100 $\times 10^{-8}$; <code>grad_bias</code> 1.676 $\times 10^{-8}$	<code>grad_hidden</code> 0.99999999999998178; <code>grad_weight</code> 0.99999999999999601; <code>grad_bias</code> 0.99999999999999991
Corrected Stage 2 retained-site replay	True k2 pruned retained sites with atomic fallback; $C = 4336$	R1 fixture references and atomic fallback adapter reference	Forward: valid/masked target log-prob, blank log-prob, log Z . Backward: cached-hidden gradient; <code>grad_weight</code> ; <code>grad_bias</code> .	Forward all 7.62939453125 $\times 10^{-8}$; backward 0.0; 3.259629011154175 $\times 10^{-9}$; 1.862645149230957 $\times 10^{-9}$	Backward 0.0; 8.776667 $\times 10^{-8}$; 8.298761 $\times 10^{-8}$	Backward 1.0; 1.0; 1.0
Stage 3 one-batch recipe-native smoke	One-batch recipe-native retained-site bridge; $C = 4336$	Compact-DP reference, retained full-logits fixture baseline, and recipe-native bridge/scatter reference	Loss vs compact reference; forward target/blank/log Z ; AMLM bridge gradients; output-head weight/bias gradients.	Loss 0.0; forward all 7.62939453125 $\times 10^{-8}$; AMLM 0.0; output weight 3.72520298461914 $\times 10^{-9}$; output bias 1.862645149230957 $\times 10^{-9}$	AMLM 0.0; output weight 8.574942 $\times 10^{-8}$; output bias 8.156776 $\times 10^{-4}$	AMLM 1.0; output weight 0.99999994039535522; output bias 1.0000001192092896

Panel B: zero rules, dtype/tolerance, status, and caveats.				
Scope / artifact	Zero-rule count if applicable	Dtype / tolerance	Status	Caveat
Large- C compact-DP boundary	Invalid target site count = 74115; invalid/inactive nonzero counts not reported in this artifact	fp32 CUDA path; row dtype not separately stored; backward pass threshold is cosine ≥ 0.999	Pass for backward parity; forward not available	Backward parity only. Finite checks are sanity only, not reference-equivalence metrics; the reference is an atomic selected-normalizer baseline rather than a dense materialized full-softmax oracle for every large C .
Cached-hidden selected-normalizer boundary	Invalid target site count = 3500	<code>torch.float32</code> ; forward max-abs threshold $\leq 5 \times 10^{-4}$; backward cosine threshold ≥ 0.999	Pass	Synthetic cached-hidden fixture; not training-loop evidence.
Cached-hidden backend-adapter boundary	Invalid target site count = 3500; inactive nonzero counts not stored for this synthetic probe	<code>torch.float32</code> ; forward max-abs threshold $\leq 5 \times 10^{-4}$; backward cosine threshold ≥ 0.999	Pass	Adapter-boundary evidence only. The older $C = 4336$ optimized-candidate interpretation is superseded by corrected Stage 2; feature-gradient accounting is diagnostic.
Synthetic multi-selected second-instance parity	Invalid target sites = 5; invalid target omitted while selected set still contains primary blank plus at least two extras	<code>torch.float32</code> ; deterministic synthetic fixture; memory scope is per- C child process after fixture tensors are allocated	Pass for $ S_i = 3..5$ selected-set parity at $C = 4336, 8192$	Synthetic second-instance parity only. PANDA-style peak memory max is 44.21/56.00 MiB PyTorch allocated/reserved after fixture allocation; dense oracle max is 77.11/90.00 MiB. This is not production multi-blank/TDT support, training, decode, WER/CER, ASR-quality, speedup, production/default-backend, multi-hardware, or absolute-firstness evidence.
Corrected Stage 2 retained-site replay	Invalid dtarget nonzero = 0; inactive dtarget nonzero = 0; inactive dblank nonzero = 0	<code>torch.float32</code> in the R1 tensor manifest; forward max-abs threshold $\leq 5 \times 10^{-4}$; backward cosine threshold ≥ 0.999	Pass for atomic fallback semantics	The optimized candidate was not run because $C < 8192$; this row is fallback correctness evidence, not optimized-candidate, training, decode, or speed evidence.
Stage 3 one-batch recipe-native smoke	Invalid dtarget nonzero = 0; inactive dtarget nonzero = 0; inactive dblank nonzero = 0	<code>torch.float32</code> fixture tensors; <code>torch 2.9.1+cu128</code> environment; backward cosine threshold ≥ 0.999	Pass for one-batch <code>compute_loss</code> to <code>loss.backward()</code> contract	One-batch plumbing evidence only: no optimizer step, training, decode, WER/CER, or ASR-quality claim. The retained full-logits baseline is a reference/oracle and may materialize retained full logits.

Table 7. Matched exact full- C integrated-path systems context rows. Scope is `integrated_path` for every row. All rows come from the matched rerun artifact using the same parent harness, environment stack, row process policy, and R1 mature AISHELL r20 fixture; memory is PyTorch peak allocated/reserved MiB. The matched rerun used `panda-k2-cul28` plus the artifact-local numba CUDA overlay. Timing is scoped systems timing, not `training_loop` timing and not a PANDA speedup claim. Loss values are finite sanity checks only, not cross-backend loss parity. OOM memory values are failure-attempt peaks; external OOM rows mark systems/OOM boundaries, not method-negative results. Previous copied R1 and RelatedWork overlay timings are excluded from the timing columns here and should be read only as `cross_artifact_same_scope_context`. The matched rerun did not attempt $C = 16384$ because three full-logits rows already OOM at $C = 8192$.

Path	Group	C	Status	Time ms	PyTorch peak allocated/reserved MiB	Persistent full logits / grad	Exact full- C / approx. denom.	Finite sanity / reading
guarded k2_mi / PANDA	PANDA context	4,336	ok	1103.620	5882.887/7696.000	No / No	true / false	finite loss; selected-normalization boundary
Materialized full logits	direct mechanism baseline	4,336	ok	658.404	13241.281/13814.000	Yes (eff.) / Yes (eff.)	true / false	finite loss; persistent full-logits context
selected.torch	implementation ablation	4,336	ok	1512.002	9105.910/11536.000	No / No	true / false	finite loss; PyTorch-level selected normalization
NeMo bottom RNNLossNumba	external practical	4,336	ok	1409.377	10539.937/11104.000	Yes (eff.) / Yes (eff.)	true / false	finite loss/grad sanity
warprnt.numba	external practical	4,336	ok	1450.972	11987.935/12552.000	Yes (eff.) / Yes (eff.)	true / false	finite loss/grad sanity
official SpeechBrain TransducerLoss	external practical	4,336	ok	1065.400	11987.935/12552.000	Yes (eff.) / Yes (eff.)	true / false	finite loss/grad sanity
guarded k2_mi / PANDA	PANDA context	8,192	ok	839.633	5895.108/7740.000	No / No	true / false	finite loss; selected-normalization boundary
Materialized full logits	direct mechanism baseline	8,192	OOM	–	14211.709/14750.000	Yes (eff.) / Yes (eff.)	true / false	systems/OOM boundary row
selected.torch	implementation ablation	8,192	ok	1753.615	11678.548/13622.000	No / No	true / false	finite loss; PyTorch-level selected normalization
NeMo bottom RNNLossNumba	external practical	8,192	ok	1387.866	14413.584/15130.000	Yes (eff.) / Yes (eff.)	true / false	finite loss/grad sanity
warprnt.numba	external practical	8,192	OOM	–	14218.528/14750.000	Yes (eff.) / Yes (eff.)	true / false	systems/OOM boundary row
official SpeechBrain TransducerLoss	external practical	8,192	OOM	–	14219.200/14750.000	Yes (eff.) / Yes (eff.)	true / false	systems/OOM boundary row

Table 8. Secondary WenetSpeech $C = 5537$ bounded real-ASR route. Scope is `training_loop`. The route is WenetSpeech Zipformer RNN-T/transducer with `blank_id=0`. The output head has weight [5537, 512] and bias [5537]; token-table rows including disambiguation are 5539, but the output class count is $C = 5537$. Memory is PyTorch peak allocated/reserved MiB. This is a bounded subset, not synthetic- C , not full training, not formal WER/CER preservation, and not ASR-quality evidence.

Path	Scope	C	Steps	Mean/median ms	PyTorch peak allocated/reserved MiB	OOM/NaN	Full logits	Full grad logits	Safe reading
Materialized	training_loop	5,537	30/30	426.431/401.627	6040.212/7348.000	0/0	Yes	Yes (eff.)	bounded WenetSpeech $C=5537$ training_loop subset; materialized full logits
guarded k2_mi / PANDA	training_loop	5,537	30/30	293.733/292.593	2119.542/2572.000	0/0	No	No	bounded WenetSpeech $C=5537$ training_loop subset; guarded k2_mi avoids persistent full logits/full grad logits

Loss drift max/final abs 0.011298/0.011298 and rel 0.004298/0.001277; final parameter rel- L_2 is $5.109e - 08$. Decode sanity status is ok, but it is not a formal WER/CER preservation result. This partially addresses the real-ASR-route coverage concern beyond AISHELL $C = 4336$ and does not close the $C \geq 8192$ concern.

The caveats are part of the result. The candidate is explicit opt-in and non-default, and the atomic fallback remains available. The 500-step gate is `fixed_fixture_replay`, not a true multi-batch recipe-native stream. No decode, WER/CER, ASR-quality, full-training, convergence, production/default-backend, benchmark/speedup, multi-hardware, or user-ready plugin claim is made.

C CURRENT NON-CLAIMS

The generated inventory table is included to keep non-claims visible in the source package. In particular, sampled-softmax rows use `K_sample` and an approximate denominator, inventory-only rows have no local metric in the inventory table, and synthetic- C rows are systems rows rather than ASR-quality rows. Practical exact full- C external `integrated_path` rows, where runnable, are reported separately in Table 7.

REFERENCES

- An, K., Shi, X., and Zhang, S. BAT: Boundary aware transducer for memory-efficient and low-latency ASR. In *Interspeech 2023*, pp. 4963–4967, 2023. doi: 10.21437/Interspeech.2023-770.
- Braun, S., McDermott, E., and Hsiao, R. Neural transducer training: Reduced memory consumption with sample-wise computation. *arXiv preprint arXiv:2211.16270*, 2023.
- Dao, T., Fu, D. Y., Ermon, S., Rudra, A., and Ré, C. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems*, volume 35, pp. 16344–16359, 2022.
- Graves, A. Sequence transduction with recurrent neural networks. *arXiv preprint arXiv:1211.3711*, 2012.

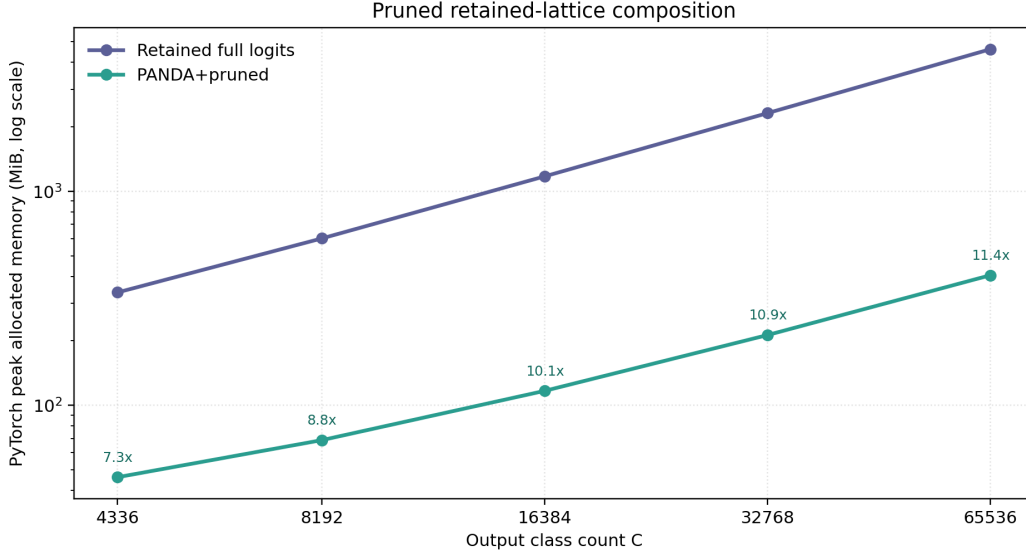


Figure 7. **Appendix visualization of pruned retained-site composition.** Scope is `integrated_path`. The y-axis is PyTorch peak allocated memory in MiB on a log scale; reserved MiB values are reported in Table 4. Pruning reduces retained lattice sites N , while PANDA reduces per-site C materialization. Larger- C rows are synthetic systems/OOM evidence, not ASR-quality evidence.

Table 9. **Appendix-only caveat evidence.** These rows are not used for the main exact full- C claim. Sampled-softmax uses `K_sample` and an approximate denominator; the PANDA-sampled plugin changes the objective in sampled mode; ESPnet BPE5000 (Watanabe et al., 2018) is portability evidence at $C = 5000$ without formal WER.

Evidence	Evidence scope(s)	Exact full- C	Approx. denom.	Safe appendix reading
Sampled-softmax H1 bounded comparison	<code>training_loop</code>	No	Yes	approximate-denominator related work; use <code>K_sample</code> for sampled class count and do not infer large- C CER/WER behavior
PANDA-sampled plugin	<code>integrated_path</code> ; <code>training_loop</code>	No in sampled mode	Yes	design observation that selected-normalizer machinery can implement a sampled objective; no plugin CER row and no main result claim
ESPnet LibriSpeech BPE5000 route	<code>training_loop</code> bounded; fixture smoke	Yes	No	portability at $C = 5000$ in bounded and fixture-smoke artifacts only; no formal WER and no main ASR-quality claim

Hsu, P.-L., Dai, Y., Kothapalli, V., Song, Q., Tang, S., Zhu, S., Shimizu, S., Sahni, S., Ning, H., and Chen, Y. Liger kernel: Efficient Triton kernels for LLM training. *arXiv preprint arXiv:2410.10989*, 2024.

Kuang, F., Guo, L., Kang, W., Lin, L., Luo, M., Yao, Z., and Povey, D. Pruned RNN-T for fast, memory-efficient ASR training. In *Interspeech 2022*, pp. 2068–2072, 2022. doi: 10.21437/Interspeech.2022-10340.

Lee, J., Lee, L., and Watanabe, S. Memory-efficient training of RNN-Transducer with sampled softmax. In *Interspeech 2022*, pp. 4441–4445, 2022. doi: 10.21437/Interspeech.2022-787.

Milakov, M. and Gimelshein, N. Online normalizer calculation for softmax. *arXiv preprint arXiv:1805.02867*, 2018.

Sudo, Y., Shakeel, M., Fukumoto, Y., Yan, B., Shi, J., Peng, Y., and Watanabe, S. Joint beam search integrating CTC, attention, and transducer decoders. *arXiv preprint arXiv:2406.02950*, 2024.

Tillet, P., Kung, H. T., and Cox, D. Triton: An intermediate language and compiler for tiled neural network computations. In *Proceedings of the 3rd ACM SIGPLAN International Workshop on Machine Learning and Programming Languages*, pp. 10–19, 2019.

Tjandra, A., Singhal, N., Zhang, D., Kalinli, O., Mohamed, A., Le, D., and Seltzer, M. L. Massively multilingual ASR on 70 languages: Tokenization, architecture, and generalization capabilities. In *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 1–5, 2023.

Watanabe, S., Hori, T., Karita, S., Hayashi, T., Nishitoba, J., Unno, Y., Soplin, N. E. Y., Heymann, J., Wiesner, M., Chen, N., Renduchintala, A., and Ochiai, T. ESPnet: End-to-end speech processing toolkit. In *Interspeech 2018*, pp. 2207–2211, 2018. doi: 10.21437/Interspeech.2018-1456.

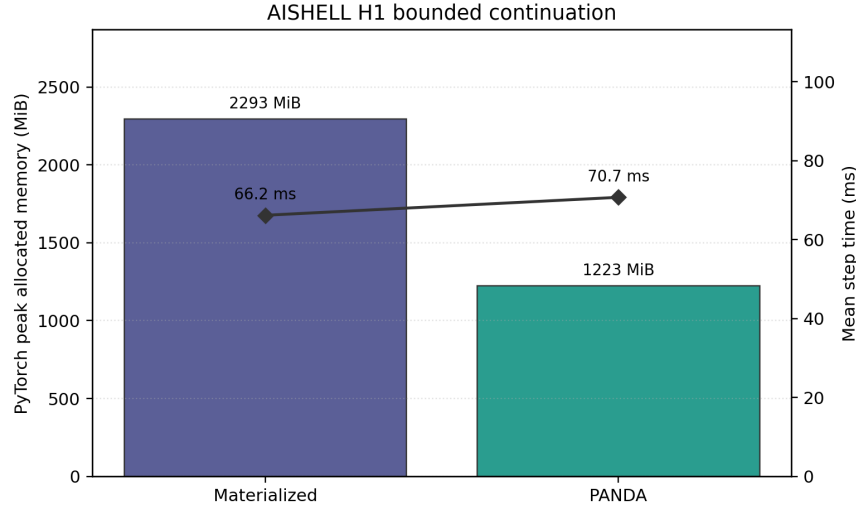


Figure 8. **Appendix visualization of the AISHELL H1 bounded continuation.** Scope is `training_loop`. Bars show PyTorch peak allocated memory in MiB; reserved MiB values and bounded CER sanity values are reported in Table 3. The line shows observed mean step time only for this bounded continuation and is not a full-training speedup claim. This real-route row is not a synthetic-*C* row; synthetic-*C* rows in this paper are systems/OOM evidence, not ASR-quality evidence.

Wijmans, E., Huval, B., Hertzberg, A., Koltun, V., and Krähenbühl, P. Cut your losses in large-vocabulary language models. In *International Conference on Learning Representations*, 2025.

Yang, Y.-Y., Hira, M., Ni, Z., Chourdia, A., Astafurov, A., Chen, C., Yeh, C.-F., Puhersch, C., Pollack, D., Genzel, D., Greenberg, D., Yang, E. Z., Lian, J., Mahadeokar, J., Hwang, J., Chen, J., Goldsborough, P., Roy, P., Narenthiran, S., Watanabe, S., Chintala, S., Quenneville-Bélair, V., and Shi, Y. TorchAudio: Building blocks for audio and speech processing. *arXiv preprint arXiv:2110.15018*, 2021.

Zhang, B., Lv, H., Guo, P., Shao, Q., Yang, C., Xie, L., Xu, X., Bu, H., Chen, X., Zeng, C., Wu, D., and Peng, Z. WenetSpeech: A 10000+ hours multi-domain mandarin corpus for speech recognition. *arXiv preprint arXiv:2110.03370*, 2022.

Zhang, Q., Lu, H., Sak, H., Tripathi, A., McDermott, E., Koo, S., and Kumar, S. Transformer transducer: A streamable speech recognition model with transformer encoders and RNN-T loss. *arXiv preprint arXiv:2002.02562*, 2020.

Table 11. Non-default retained-site adapter prototype evidence. These rows are appendix-only engineering-extension evidence. They document an explicit opt-in retained-site adapter direction for the AISHELL pruned2 retained-site route at $C = 4336$; they are not a production/default backend, not a benchmark, not full training, and not ASR-quality evidence. Memory is PyTorch peak allocated/reserved MiB when reported.

Evidence	Scope	Key result	Safe reading
Adapter API contract	API contract	Pass; layout is <code>true_k2_pruned_retained_sites</code> with $N = 3460$ flat sites, 2260 active sites, $C = 4336$, and <code>blank_id=0</code> .	Defines a narrow explicit opt-in shape gate for the grouped candidate; atomic fallback remains available.
Real retained-site fixture adapter	<code>real_retained_site_adapter</code> boundary	Atomic adapter and shape-gated candidate both pass fixture/layout validation; peak allocated/reserved is 87.007/122.000 MiB for both rows.	Real retained-site fixture validation only; no training, decode, production/default backend, or speedup claim.
Recipe-native loss/backward	one-batch <code>one_batch_loss/backward</code> smoke	Atomic and explicit opt-in candidate both report loss 0.026498144 and complete <code>loss.backward()</code> .	One-batch loss/backward integration smoke only; no optimizer step, training loop, decode, WER/CER, or benchmark claim.
Optimizer-step and short smoke	<code>one_batch_optimizer-step</code> smoke; 20-step short smoke	Atomic and explicit opt-in candidate pass 1/1 optimizer step and 20/20 short smoke steps with finite gradients and no OOM/NaN.	Short smoke/integration evidence only; timing is not benchmark or speedup evidence.
Staged fixed-fixture replay	<code>fixed_fixture_replay</code> , 500-step bounded gate	Atomic and explicit opt-in candidate both complete 500/500 steps. Loss first/final/min/max is 0.026491666 / 0.026104022 / 0.026104022 / 0.026491666; peak allocated/reserved is 188.411/262.000 MiB for both rows.	Fixed-fixture replay only, not a true multi-batch recipe-native stream. Both rows keep both persistence flags false.